

HTML (HyperText Markup Language)

What is HTML?

HTML (HyperText Markup Language) is the standard language used to create and structure content on the web. It uses tags to tell the browser what type of content to display headings, paragraphs, images, links, forms, and more.

HTML is not a programming language it is a markup language. HTML5 is the current version and introduced semantic elements such as <header>, <main>, <section>, and <footer> that give meaningful structure to a page.

HTML Document Structure

Every HTML page follows a standard boilerplate structure. This is the template used across all pages in this portfolio:

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Page Title</title>
    <link rel="stylesheet" href="style.css" />
  </head>
  <body>

    <header>
      <nav><!-- navigation links --></nav>
    </header>

    <main>
      <section>
        <h1>Welcome</h1>
        <p>This is a paragraph.</p>
      </section>
    </main>

    <footer>
      <p>&copy; 2025 Raunak Poudel</p>
    </footer>

  </body>
</html>
```

Key parts of the structure:

- **<!DOCTYPE html>** : tells the browser this is an HTML5 document.
- **<html lang="en">** : root element; lang sets the language for accessibility.
- **<head>** : contains metadata, page title, and links to CSS/JS files.
- **<body>** : contains all visible page content rendered in the browser.

Common HTML Tags

Tag	Purpose	Example
<h1> to <h6>	Headings	Main Title
<p>	Paragraph	Some text here.
<a>	Hyperlink	Click
	Image	
/	Lists	Item
<table>	Table	<table><tr><td>Data</td></tr></table>
<form>	Form	<form action="/submit" method="POST">
<div>	Block container	<div class="card">...</div>
	Inline container	text

Code Example: HTML Form

Forms collect data from the user and send it to a server (PHP). The action attribute specifies the PHP file that handles the data. method="POST" sends data securely in the request body.

```
<form action="process.php" method="POST">

  <label for="name">Full Name</label>
  <input type="text" id="name" name="name" required />

  <label for="email">Email Address</label>
  <input type="email" id="email" name="email" required />

  <label for="message">Message</label>
  <textarea id="message" name="message" rows="4"></textarea>

  <button type="submit">Send Message</button>

</form>
```

What each part does:

- **name attribute** : PHP reads this via \$_POST['name'], \$_POST['email'], etc.
- **type="email"** : browser validates the format before submission.
- **required** : prevents submission if the field is empty.

Code Example: HTML Table

Tables display structured data in rows and columns. The <thead> contains header rows and <tbody> contains the data rows.

```
<table>
  <thead>
    <tr>
      <th>Name</th>
      <th>Subject</th>
      <th>Grade</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Raunak Poudel</td>
      <td>CET138</td>
```

```
<td>A</td>
</tr>
<tr>
  <td>Student 2</td>
  <td>CET138</td>
  <td>B</td>
</tr>
</tbody>
</table>
```

Semantic HTML5 Elements

Semantic elements describe the meaning of the content they contain, making pages more accessible and easier for search engines to understand.

- **<header>** : introductory content or navigation bar at the top.
 - **<nav>** : a block of navigation links.
 - **<main>** : the primary unique content of the page.
 - **<section>** : a thematic grouping of content with a heading.
 - **<article>** : self-contained content that could stand alone (e.g. a blog post).
 - **<footer>** : closing content such as copyright and contact information.
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